40-question answer key and explanation guide

REVIEW DRAFT. Source-authored explanations retained; independent academic validation is pending.

01-B 02-E 03-A 04-B 05-E 06-B 07-D 08-A
09-D 10-E 11-A 12-C 13-C 14-D 15-C 16-B
17-C 18-E 19-D 20-A 21-B 22-A 23-C 24-C
25-A 26-B 27-B 28-D 29-E 30-A 31-D 32-D
33-C 34-E 35-E 36-B 37-A 38-C 39-D 40-B

1. Why does the sinoatrial (SA) node normally determine the heart rate?

Answer: B. It reaches the threshold for spontaneous discharge sooner than other pacemakers.

Explanation: The SA node has the highest normal intrinsic pacemaker rate. Its impulses reach subsidiary pacemakers before those slower sites discharge.

Other options: Blood supply, contraction strength and ventricular conduction speed do not set the normal pacemaker hierarchy. Other conducting tissues also show automaticity.

Source: DOM-CVS-0001 | CIRCULATION 1, pp. 7-9

2. Left ventricular pressure has risen to 45 mmHg. Left atrial pressure is 10 mmHg and aortic pressure is 85 mmHg. Which phase is most likely?

Answer: E. Isovolumetric contraction

Explanation: The mitral valve is closed because ventricular pressure exceeds atrial pressure. The aortic valve remains closed because ventricular pressure is still below aortic pressure. Rising pressure with both valves closed indicates isovolumetric contraction.

Other options: Ejection requires ventricular pressure above aortic pressure, while filling requires ventricular pressure below left atrial pressure.

Source: DOM-CVS-0003 | CIRCULATION 1, pp. 13-15, 21

3. The diagram shows left ventricular pressure (solid) and aortic pressure (dashed). At which labelled point does aortic valve opening begin?

Answer: A. Point B

Explanation: The aortic valve opens as rising left ventricular pressure first exceeds aortic pressure. On the plotted curves this happens at B. Point A precedes the pressure crossover; C and D occur later in ejection, and E follows valve closure.

Other options: All other labels are placed before or after the initial ventricular-aortic pressure crossover.

Source: DOM-CVS-0005 | CIRCULATION 1, pp. 14-16, 21

4. A ventricle has an end-diastolic volume of 140 mL and an end-systolic volume of 56 mL. What is its ejection fraction?

Answer: B. 60%

Explanation: Stroke volume = 140 - 56 = 84 mL. Ejection fraction = (84/140) x 100 = 60%.

Other options: 84 mL is the stroke volume, not a percentage; 40% is the fraction remaining after systole.

Source: DOM-CVS-0006 | CIRCULATION 1, p. 16

5. In complete AV dissociation, an atrium sometimes contracts while the tricuspid valve is closed. Which jugular venous finding may appear?

Answer: E. Intermittent giant a waves

Explanation: Atrial contraction against a closed tricuspid valve causes a sudden large rise in right atrial pressure, producing cannon or giant a waves.

Other options: The a wave reflects atrial contraction; the v wave mainly reflects atrial filling while the tricuspid valve is closed.

Source: DOM-CVS-0008 | CIRCULATION 1, pp. 24-26

6. Why does inspiration normally widen the split between the aortic and pulmonary components of the second heart sound?

Answer: B. Increased right ventricular filling delays pulmonary valve closure.

Explanation: Inspiration increases venous return to the right heart, often prolonging right ventricular ejection and delaying P2 relative to A2.

Other options: S2 reflects semilunar valve closure, not AV valve opening or closure.

Source: DOM-CVS-0009 | CIRCULATION 1, pp. 18-19, 27-30

7. A patient has backward blood flow from the left ventricle into the left atrium during systole. Which valve disorder best explains this?

Answer: D. Mitral regurgitation

Explanation: Mitral regurgitation occurs when the mitral valve fails to prevent systolic backflow from the left ventricle into the left atrium.

Other options: Mitral stenosis obstructs diastolic forward flow; aortic disorders affect flow across the aortic valve, and tricuspid regurgitation affects the right heart.

Source: DOM-CVS-0010 | CIRCULATION 1, pp. 31-33

8. Venous return rises while heart rate, afterload and contractility remain unchanged. What immediate change is expected in a normal ventricle?

Answer: A. A rise in end-diastolic volume and stroke volume

Explanation: More venous return increases ventricular filling and stretches cardiac fibres. On the ascending Frank-Starling relationship this raises force and stroke volume.

Other options: The Frank-Starling response links greater preload to greater output within physiological limits.

Source: DOM-CVS-0012 | CIRCULATION 1, pp. 38-44

9. Aortic pressure suddenly rises. For the first affected beat, ventricular filling and contractility are unchanged. What is the likely effect on stroke volume?

Answer: D. Stroke volume decreases because the ventricle ejects against greater pressure.

Explanation: Greater afterload makes ventricular ejection more difficult. Before compensatory adjustments, end-systolic volume tends to increase and stroke volume decreases.

Other options: Aortic recoil, filling and AV conduction are not the mechanisms determining the first-beat afterload response.

Source: DOM-CVS-0013 | CIRCULATION 1, pp. 38, 45-46

10. Two ventricular function curves are shown. At the same end-diastolic volume, curve X produces a larger stroke volume than curve Y. What best explains the difference?

Answer: E. Ventricular contractility is greater on curve X.

Explanation: An upward shift of the Frank-Starling curve indicates greater stroke volume at the same preload, consistent with increased contractility.

Other options: The comparison holds end-diastolic volume constant, so increased preload is not the explanation; reduced calcium response would tend to reduce contractility.

Source: DOM-CVS-0014 | CIRCULATION 1, pp. 39-41, 45-47

11. Oxygen consumption is 280 mL/min. Arterial oxygen content is 200 mL/L and mixed venous oxygen content is 130 mL/L. What is cardiac output?

Answer: A. 4 L/min

Explanation: Arterial-mixed venous oxygen content difference = 200 - 130 = 70 mL/L. Cardiac output = 280 mL/min divided by 70 mL/L = 4 L/min.

Other options: The Fick method divides oxygen consumption by the oxygen content difference, not by either concentration alone.

Source: DOM-CVS-0016 | CIRCULATION 1, pp. 52-53

12. At the same stroke volume and heart rate, which change generally increases myocardial oxygen demand more?

Answer: C. Pumping against a much higher arterial pressure

Explanation: Pressure work, which increases with afterload and myocardial wall tension, carries a high oxygen cost. Pumping the same volume against greater arterial pressure is more demanding.

Other options: The alternative changes reduce pressure load rather than increase it.

Source: DOM-CVS-0018 | CIRCULATION 1, p. 54

13. According to Poiseuille's law, if a vessel radius is reduced to half while other factors remain constant, how does resistance change?

Answer: C. It becomes 16 times greater.

Explanation: Resistance varies inversely with the fourth power of radius. Halving radius therefore increases resistance by 2^4 = 16 times.

Other options: The fourth-power dependence is stronger than a linear, square or cubic change.

Source: DOM-CVS-0019 | CIRCULATION 1, pp. 68-70

14. Why is average blood-flow velocity normally much lower in capillaries than in the aorta?

Answer: D. The combined cross-sectional area of all capillaries is much larger.

Explanation: For a given volume flow rate, average velocity equals flow divided by cross-sectional area. The enormous total capillary area produces low average velocity.

Other options: Flow continues in capillaries; the key comparison is total cross-sectional area, not the diameter of one capillary.

Source: DOM-CVS-0020 | CIRCULATION 1, pp. 61, 67

15. Which isolated change increases the tendency for turbulent flow in a blood vessel?

Answer: C. An increase in blood-flow velocity

Explanation: Reynolds number increases with velocity, density and diameter, and decreases with viscosity. Increased velocity therefore makes turbulence more likely.

Other options: Higher viscosity and smaller diameter generally lower Reynolds number when other variables are held constant.

Source: DOM-CVS-0022 | CIRCULATION 1, pp. 64-66

16. A resting adult has a blood pressure of 120/75 mmHg. Using the usual resting approximation, what is mean arterial pressure?

Answer: B. 90 mmHg

Explanation: Pulse pressure = 120 - 75 = 45 mmHg. MAP is approximately diastolic pressure + one-third of pulse pressure = 75 + 15 = 90 mmHg.

Other options: MAP is not the simple arithmetic midpoint; diastole usually occupies more time at rest.

Source: DOM-CVS-0024 | CIRCULATION 1, pp. 79-82

17. In a healthy adult, cuff pressure is slowly released during blood-pressure measurement. Disappearance of Korotkoff sounds usually marks which pressure?

Answer: C. Diastolic arterial pressure

Explanation: As cuff pressure falls below diastolic pressure, the brachial artery is open throughout the cardiac cycle and the characteristic Korotkoff sounds generally disappear.

Other options: The first appearance of sound usually marks systolic rather than diastolic pressure.

Source: DOM-CVS-0025 | CIRCULATION 1, pp. 79-81

18. A deep vein in one leg becomes obstructed, increasing pressure in vessels upstream. What local change most directly favours swelling?

Answer: E. An increase in capillary hydrostatic pressure

Explanation: Impaired venous drainage raises capillary pressure and promotes movement of fluid from capillaries into interstitial tissue.

Other options: Higher plasma oncotic pressure or stronger lymphatic removal would oppose fluid accumulation.

Source: DOM-CVS-0027 | CIRCULATION 1, pp. 85-86, 90-93

19. A patient develops chronic swelling of an arm after axillary lymph nodes are removed. What is the main reason?

Answer: D. Interstitial fluid and proteins cannot drain normally through lymph vessels.

Explanation: Damage to lymphatic pathways reduces the return of interstitial fluid and proteins to the blood, producing lymphoedema.

Other options: Lymphoedema is caused by impaired lymph removal, not complete loss of arterial flow.

Source: DOM-CVS-0029 | CIRCULATION 1, pp. 60, 94-95; Regional, pp. 47-48

20. Why does walking help return blood from the leg veins to the heart?

Answer: A. Muscle contractions compress veins while valves limit backward flow.

Explanation: Contraction of leg muscles compresses nearby veins. One-way venous valves direct the displaced blood centrally, limiting pooling.

Other options: Venous valves are passive one-way structures; muscle compression supplies the pumping action.

Source: DOM-CVS-0030 | CIRCULATION 1, pp. 97-98

21. During exercise, working skeletal muscle produces more CO2, H+ and other metabolites. What local vascular response is expected?

Answer: B. Arteriolar dilation that increases blood supply to the active muscle

Explanation: Local metabolic vasodilators help match perfusion to greater tissue metabolic activity. Dilation reduces resistance and permits more flow.

Other options: Active tissue usually needs greater, not lower, local perfusion.

Source: DOM-CVS-0031 | CIRCULATION 2, pp. 4-6

22. Blood flow through a forearm artery is briefly blocked and then restored. Why may flow temporarily rise above its resting level?

Answer: A. Vasodilator metabolites built up while the tissue was underperfused.

Explanation: During the occlusion, tissue oxygen delivery falls and local vasodilatory signals build up. When flow resumes, dilated resistance vessels produce transient reactive hyperaemia.

Other options: Reactive hyperaemia is a temporary local response to preceding flow interruption.

Source: DOM-CVS-0033 | CIRCULATION 2, pp. 5-6, 11

23. The graph shows blood flow across a range of perfusion pressures. Between points B and C, why is flow nearly constant despite rising pressure?

Answer: C. Arteriolar resistance rises as perfusion pressure rises.

Explanation: For approximately constant flow Q, a higher perfusion pressure requires a higher local resistance (Q = pressure difference / resistance). The plateau reflects autoregulatory adjustment.

Other options: Falling resistance alongside increasing pressure would increase, rather than stabilize, flow.

Source: DOM-CVS-0034 | CIRCULATION 2, pp. 4-10

24. A person suddenly loses blood and arterial pressure falls. Which immediate autonomic change helps restore pressure?

Answer: C. Increased sympathetic activity and reduced vagal activity

Explanation: Less stretch of carotid and aortic baroreceptors reduces afferent firing. The medulla responds with increased sympathetic and reduced parasympathetic outflow.

Other options: Increased vagal activity would tend to slow the heart rather than compensate for falling pressure.

Source: DOM-CVS-0035 | CIRCULATION 2, pp. 18-27

25. A patient develops rapidly rising intracranial pressure with impaired brainstem perfusion. Which combination is typical of the Cushing response?

Answer: A. Increased arterial pressure with reflex slowing of the heart

Explanation: Reduced medullary perfusion causes a powerful sympathetic rise in systemic pressure. The resulting arterial hypertension stimulates baroreceptors, producing reflex bradycardia.

Other options: The raised pressure is a compensatory response to brainstem ischemia; bradycardia is secondary to the baroreflex.

Source: DOM-CVS-0037 | CIRCULATION 2, pp. 31-36; Regional, pp. 22-23

26. Rapid infusion of fluid increases venous return and stretches the right atrium. Which reflex response may help accommodate the larger venous return?

Answer: B. Heart rate may increase as atrial stretch activates a reflex.

Explanation: The atrial stretch (Bainbridge) reflex can increase heart rate when venous return and atrial filling rise. It is distinct from the arterial baroreceptor response to elevated blood pressure.

Other options: Atrial distension can elicit reflex tachycardia without requiring arterial hypotension. The infusion increases rather than reduces venous return.

Source: DOM-CVS-0038 | CIRCULATION 2, pp. 28-29

27. Immediately after a person stands up quickly, before reflex compensation has fully developed, what tends to happen?

Answer: B. Venous pooling increases and venous return to the heart falls.

Explanation: Gravity increases hydrostatic pressure and pooling in dependent veins. This reduces central venous return, initially lowering stroke volume and potentially cerebral perfusion.

Other options: Without compensation, standing does not drive leg blood centrally or raise head-level perfusion.

Source: DOM-CVS-0039 | Homeostasis, pp. 2-8

28. A person loses a substantial amount of blood. What sequence most directly explains the initial fall in cardiac output?

Answer: D. Lower blood volume leads to lower venous return and ventricular filling.

Explanation: Blood loss reduces circulating volume and venous return, which reduces preload and stroke volume. Reflex vasoconstriction may defend pressure but cannot automatically restore lost flow.

Other options: Loss of circulating volume does not directly raise preload or stroke volume.

Source: DOM-CVS-0041 | Homeostasis, pp. 12-20

29. A patient with severe infection has widespread arteriolar dilation and low arterial pressure. Which shock mechanism is most likely?

Answer: E. Distributive shock caused mainly by reduced systemic vascular resistance

Explanation: Sepsis can produce profound vasodilation and loss of vascular tone, lowering total peripheral resistance and effective perfusion pressure. This is distributive shock.

Other options: The defining event in this stem is widespread low resistance rather than haemorrhage, pump failure or obstruction.

Source: DOM-CVS-0042 | Homeostasis, pp. 13-16, 28-29

30. A patient with left ventricular failure develops breathlessness and pulmonary oedema. What pressure change most directly promotes fluid movement into the lungs?

Answer: A. Increased pulmonary venous and capillary hydrostatic pressure

Explanation: Poor left ventricular output raises pressure upstream in the left atrium and pulmonary veins. Higher pulmonary capillary hydrostatic pressure promotes filtration and pulmonary oedema.

Other options: Fluid filtration increases with raised hydrostatic pressure; higher oncotic pressure would tend to oppose it.

Source: DOM-CVS-0044 | Homeostasis, pp. 31-33

31. An obstruction mainly affects blood supply to the lateral surface of a cerebral hemisphere. Which major cerebral artery is most likely involved?

Answer: D. Middle cerebral artery

Explanation: The middle cerebral artery supplies much of the lateral cerebral hemisphere. The anterior cerebral artery supplies much of the medial frontal and parietal surface.

Other options: The anterior and posterior cerebral arteries supply other major territories; communicating arteries are chiefly connectors in the arterial circle.

Source: DOM-CVS-0046 | Regional Circulation, pp. 2-4

32. The cerebral aqueduct becomes blocked while CSF production continues. Which ventricles are expected to enlarge first?

Answer: D. The lateral and third ventricles

Explanation: The aqueduct links the third to the fourth ventricle. Obstruction causes CSF to accumulate upstream in the third and lateral ventricles.

Other options: The fourth ventricle lies downstream of the aqueduct and is not the primary site of enlargement in an isolated aqueduct blockage.

Source: DOM-CVS-0047 | Regional Circulation, pp. 6-9, 16

33. Mean arterial pressure is 95 mmHg and intracranial pressure is 15 mmHg. Assuming cerebral venous pressure is not higher than intracranial pressure, what is cerebral perfusion pressure?

Answer: C. 80 mmHg

Explanation: Cerebral perfusion pressure is approximately MAP - ICP when ICP is the effective downstream pressure: 95 - 15 = 80 mmHg.

Other options: Perfusion pressure is a difference between driving and downstream pressures; it is not the MAP alone.

Source: DOM-CVS-0049 | Regional Circulation, pp. 21-23

34. Why does most left ventricular coronary blood flow occur during diastole rather than systole?

Answer: E. Contracting left ventricular muscle compresses vessels within its wall.

Explanation: During left ventricular systole, high intramyocardial pressure compresses coronary vessels, especially in the subendocardium. Relaxation during diastole permits better perfusion.

Other options: Coronary arterial openings remain connected to the aorta; the critical factor is compression by contracting myocardium.

Source: DOM-CVS-0050 | Regional Circulation, pp. 26-28

35. One group of alveoli is poorly ventilated while nearby alveoli are well ventilated. What happens to the small pulmonary arteries supplying the poorly ventilated region?

Answer: E. They constrict, directing more blood toward better-ventilated alveoli.

Explanation: Low alveolar oxygen causes local pulmonary arteriolar constriction. This helps divert flow toward alveoli where oxygen uptake is more effective.

Other options: The pulmonary hypoxic response is opposite to the usual local hypoxic vasodilation in systemic tissues.

Source: DOM-CVS-0052 | Regional Circulation, pp. 31-32

36. During exercise, pulmonary blood flow increases greatly but pulmonary arterial pressure rises relatively little. What helps explain this?

Answer: B. Previously underperfused capillaries open and existing vessels widen.

Explanation: Recruitment of additional pulmonary capillaries and distension of vessels increase the total flow capacity, limiting the rise in pulmonary vascular resistance and pressure.

Other options: Recruitment and distension expand, not reduce, pulmonary vascular capacity.

Source: DOM-CVS-0053 | Regional Circulation, pp. 30-32 (concept extension)

37. After firm stroking of the skin, redness spreads beyond the directly stimulated line. What produces this surrounding flare?

Answer: A. An axon reflex causing nearby arterioles to dilate

Explanation: The flare of the triple response is produced by an antidromic axon reflex that dilates vessels around the stimulated region.

Other options: The wheal reflects increased local permeability, whereas the flare is spreading arteriolar vasodilation.

Source: DOM-CVS-0055 | Regional Circulation, pp. 39-40

38. What change occurs in the pulmonary circulation with the newborn's first breaths?

Answer: C. Pulmonary vascular resistance falls as the lungs expand and oxygen rises.

Explanation: Lung expansion and increased alveolar oxygen reduce pulmonary vascular resistance, raising pulmonary blood flow. Loss of the placenta simultaneously tends to raise systemic vascular resistance.

Other options: The first breaths promote pulmonary vasodilation, not vasoconstriction.

Source: DOM-CVS-0057 | Regional Circulation, pp. 42-45 (concept extension)

39. Which pair of electrodes forms standard bipolar lead I in an ECG?

Answer: D. Right arm (negative) and left arm (positive)

Explanation: Standard lead I measures the potential difference LA(+) minus RA(-). Lead II uses RA(-) and LL(+), while lead III uses LA(-) and LL(+).

Other options: Lead II and III use the left-leg electrode; reversing the lead I polarity is also incorrect.

Source: DOM-CVS-0058 | ECG 2, pp. 3-5

40. In the ECG strip, the distance between consecutive R waves is 20 small squares. The paper speed is 25 mm/s. What is the heart rate?

Answer: B. 75 beats/min

Explanation: At 25 mm/s, one small square = 0.04 s. Twenty squares = 0.8 s per beat, giving 60/0.8 = 75 beats/min. Equivalently, 1500/20 = 75 beats/min.

Other options: Heart rate depends on the R-R interval; the standard 25 mm/s conversion yields 1500 divided by small squares.

Source: DOM-CVS-0060 | ECG 2, pp. 12-14